1.1 The cell membrane

Structure

The cell membrane is a phospholipid bilayer. The hydrophilic heads face the water on each side, and the hydrophobic tails point inward.

Proteins are scattered through the bilayer. This arrangement is described by the fluid mosaic model.

Function

The membrane is selectively permeable: small non-polar molecules such as oxygen pass freely, while ions need channel proteins.

Movement of water across the membrane from high to low water potential is called osmosis.

Transport

Active transport moves substances against their concentration gradient and requires energy from ATP.

Facilitated diffusion uses carrier proteins to move substances down their gradient without using energy.

Large particles are taken in by endocytosis and released by exocytosis.

1.2 The nucleus

Structure

The nucleus is surrounded by a double membrane called the nuclear envelope, which is perforated by nuclear pores.

Inside, DNA is wound around proteins called histones to form chromatin.

Function

The nucleolus is the site where ribosomes are assembled.

1.3 Mitochondria

Structure

Mitochondria have an inner membrane folded into cristae, which increase the surface area.

Function

Mitochondria are the site of aerobic respiration, producing most of the cell's ATP.

They contain their own circular DNA, supporting the endosymbiotic theory.

2.1 DNA structure

The double helix

DNA is made of two strands of nucleotides twisted into a double helix.

Each nucleotide contains a deoxyribose sugar, a phosphate group and a base.

Base pairing

Adenine pairs with thymine, and guanine pairs with cytosine, joined by hydrogen bonds.

2.2 Mitosis

Stages

The four stages of mitosis are prophase, metaphase, anaphase and telophase.

In metaphase, chromosomes line up at the equator of the cell.

Result

Mitosis produces two daughter cells that are genetically identical to the parent cell.

2.3 Meiosis

Result

Meiosis produces four daughter cells, each with half the number of chromosomes.

Variation arises from crossing over and independent assortment.